Foundations of Human-Computer Interaction

Course Information

Course Number/Section	CS 4352.002/CGS 4352
Course Title	Foundations of Human-Computer Interaction
Term	Fall 2026
Days & Times	Tuesday & Thursday 11:30 AM – 12:45 PM
Location	ECSS 2.412
Course Modality	Traditional/Face-to-face

Professor Information

Instructor	Dr. Liang He, Ph.D. / www.lianghe.me
Office Phone	(412) 320-6389
Email Address	liang.he@utdallas.edu
Office Location	ECSS 4.224
Student Hours	Wed 10:00 AM – 11:00 AM + upon email request

Teaching Assistant Information

Teaching Assistant	Yili (Angel) We, Xiaoman Yang
Email Address	yili.wen@utdallas.edu, xiaoman.yang@utdallas.edu
Student Hours	Thursdays 10:15 – 11:15 AM CT at ECSS 4.217

Course Modality and Expectations

Instructional Mode	Traditional (In-Person)
Course Platform	The course will be taught face-to-face. Instructor and students meet according to the schedule.
Class Attendance	You will attend every class synchronously unless you have a valid reason, such as medical issues. ***Valid proof is required and submitted to the instructor and TA at least 24 hours before the class time via email.***

Course Description

Theory and practice of human-computer interaction (HCI). Provides an introductory exploration of human-computer interaction, focusing on the design thinking process. Topics include an iterative process of empathize, define, ideate, prototype, and test around understanding people, user research, and user interafces. A significant part of this course will be a hands-on class project for designing, prototyping, and evaluating user interfaces.

Student Learning Objectives/Outcomes

- Ability to understand the design thinking process
- Ability to develop low-fi and high-fi prototypes
- Ability to design useful and usable interfaces that are carefully catered to users' needs
- Ability to apply heuristics evaluation and test usability

Required Textbooks and Materials

There is no official textbook. We will mix various resources that best explain the core concepts in human-computer interaction. Instructor will post suggested resources/materials on Canvas.

(Optional Reference Books)

- "Interaction Design: Beyond Human-Computer Interaction" by Yvonne Rogers, Helen Sharp, and Jennifer Preece (6th Edition)
- "Introduction to Human-Computer Interaction" by Kasper Hornbæk, Per Ola Kristensson, and Antti Oulasvirta (2025)
- "Ways of Knowing in HCI" by Judith S. Olson and Wendy A. Kellogg (2014)

Textbooks and some other bookstore materials can be ordered online or purchased at the UT Dallas Bookstore.

In addition to a confident level of computer and Internet literacy, certain minimum technical requirements must be met to enable a successful learning experience. Please review the important technical requirements on the UT Dallas Canvas Migration webpage.

Grading Policy

Credit Distribution

- Design Project: 70%
  - DP0: Check-In (1%)
  - DP1: Needfinding (8%)
  - DP2: Ideation (8%)
  - DP3: Interactive Prototype (14%)
  - DP4: Implementation (22%)
  - DP5: Heuristic Evaluation (5%)
  - DP6: Final Demo (12%)
- Activities: 4%
- Paper Presentation: 6%

- Exam: 15%
- Participation: 5%

Grading Scale

- A 93 or above
- A- 90-93
- B+ 87-90
- B 83-87
- B- 80-83
- C+ 77-80
- C 73-77
- C- 70-73
- D+ 67-70
- D 63-67
- D- 60-63
- F 60 or below

Course Policies

- Design project, in-class activities, and paper presentation will be team-based.
- Canvas is the official information portal for this course. Course announcements, lecture slides, programming assignments, project, and grades will be communicated via Canvas.
- Final course grade will be posted in Galaxy by the Records Office.
- There will be 1-point deduction for each class absence in participation score (5%). There will be zero point for class participation if the number of absences is more than three. Students should consult to the instructor and TA in advance (at least 24-hours before the class time) through email and get excuses with valid evidence.
- If you decide to stop attending class, be sure to drop or withdraw from the course. Otherwise, you risk receiving an 'F' or 'NF' for the course.
- Final course grade will be posted in Galaxy by the Records Office.

Regrade Policy

- For project grading, you have a window of 7 days (from when we return your assignment) to ask for a regrade. We will not consider any regrade requests outside this window. Regrade requests should be emailed to TA; regrade requests will not be considered unless they contain a clear explanation on why a regrade should be issued.

Tentative Schedule (subject to change)

Week	Topics	Deadlines

1	8/25, 8/27 Course Overview, Empathy, Activity 1	DP0 Due (8/30)
2	9/1, 9/3 Define, Activity 2, Paper Presentation, DP1	
3	9/8, 9/10 Ideate, Activity 3, Concept Videos, DP2, RP Demo	DP1 Due (9/13)
4	9/15, 9/17 Prototype, Activity 4, DP2 Worktime	
5	9/22, 9/24 Design Project Pitch Ideation, Pitch Preparation	DP2 Due (9/27)
6	9/29, 10/1 Design Project Pitch Presentations	
7	10/6, 10/8 DP3+Wireframe, Paper Presentation 1	
8	10/13, 10/15 Heuristic Evaluation, Activity 5, DP Studio	DP3 Due (10/18)
9	10/20, 10/22 Usability Testing, Interaction & UI	
10	10/27, 10/29 DP Studio, Physical Computing & TUI	
11	11/3, 11/5 DP4 Progress Check, Paper Presentation 2	
12	11/10, 11/12 Paper Presentation 3, Digital Fabrication	DP4 Due (11/15)
13	11/17, 11/19 UI Interactive Demo	DP5 Due (11/22)
14	11/24, 11/26 Fall Break & Thanksgiving (No Classes)	
15	12/1, 12/3 Paper Presentation 4, Final Exam	
16	12/8, 12/10 Final Project Presentations	DP6 Due (12/13)

UT Dallas Syllabus Policies and Procedures

Please visit http://go.utdalls.edu/syllabus-policies for other policies.

The descriptions and timelines contained in this syllabus are subject to change at the discretion of the Professor.